

QuadSync Tech Solutions

Synchronizing Innovations With Success

AI/ML Internship (Syllabus)

Content

Sr. No	Syllabus
1.	Python for AI & ML
2.	Data Analysis using NumPy & Pandas
3.	Statistics & Mathematics for ML
4.	Machine Learning Fundamentals
5.	Supervised Learning Algorithms
6.	Unsupervised Learning
7.	Deep Learning Fundamentals
8.	Advanced AI + Final Project

Python for AI & ML:

Chapter No.	Topic	Days
Chapter 1	Python Basics (Variables, Data Types, Operators, Input/Output)	Day 1
Chapter 2	Control Statements (if-else, loops), Functions (User-defined functions, Lambda functions)	Day 2
Chapter 3	Data Structures (List, Tuple, Set, Dictionary)	Day 3
Chapter 4	File Handling, Exception Handling	Day 4
Chapter 5	OOP Concepts	Day 5

Data Analysis using NumPy & Pandas :

Chapter No.	Topic	Days
Chapter 1	NumPy	Day 6
Chapter 2	Pandas-I	Day 7
Chapter 3	Pandas-II	Day 8
Chapter 4	Data Visualization (Matplotlib, Seaborn)	Day 9
Chapter 5	Mini Project	Day 10

Statistics & Mathematics for ML :

Chapter No.	Topic	Days
Chapter 1	Statistics (Mean, Median, Mode, Variance, Standard Deviation)	Day 11
Chapter 2	Probability(Basic Probability, Conditional Probability)	Day 12
Chapter 3	Linear Algebra (Vectors, Matrices)	Day 13
Chapter 4	Calculus (Derivatives, Gradient Descent Intuition)	Day 14
Chapter 5	Data Preprocessing (Scaling, Encoding, Feature Engineering)	Day 15

Machine Learning Fundamentals:

Chapter No.	Topic	Days
Chapter 1	Introduction to ML (Supervised Learning, Unsupervised Learning, Reinforcement Learning)	Day 16
Chapter 2	ML Workflow (Data Collection, Data Cleaning, Feature Engineering, Training, Evaluation, Deployment)	Day 17
Chapter 3	Train-Test Split, Performance Metrics	Day 18
Chapter 4	Practical	Day 19
Chapter 5	Mini Project	Day 20

Supervised Learning Algorithms :

Chapter No.	Topic	Days
Chapter 1	Regression (Linear Regression, Multiple Linear Regression)	Day 21
Chapter 2	Classification (Logistic Regression, KNN)	Day 22
Chapter 3	Classification (Decision Tree, Random Forest)	Day 23
Chapter 4	Practical-I	Day 24
Chapter 5	Practical-II	Day 25

Unsupervised Learning:

Chapter No.	Topic	Days
Chapter 1	Clustering	Day 26
Chapter 2	Dimensionality Reduction	Day 27
Chapter 3	Anomaly Detection	Day 28
Chapter 4	Practical-I	Day 29
Chapter 5	Practical-II	Day 30

Deep Learning Fundamentals :

Chapter No.	Topic	Days
Chapter 1	Neural Networks	Day 31
Chapter 2	Deep Learning Concepts	Day 32
Chapter 3	Framework-I TensorFlow	Day 33
Chapter 4	Framework-II Keras	Day 34
Chapter 5	Mini Project	Day 35

Advanced AI + Final Project :

Chapter No.	Topic	Days
Chapter 1	Computer Vision	Day 36
Chapter 2	NLP	Day 37
Chapter 3	Generative AI	Day 38
Chapter 4	Model Deployment	Day 39
Chapter 5	Final Capstone Project Presentation	Day 40